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Influence of solvent addition on the physicochemical properties of Brazilian gasoline

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Abstract

The influence of several solvents (anhydrous ethanol, white spirit, alkylbenzene AB9, diesel) on the physicochemical parameters of gasoline was studied according to ASTM international standard methods. The parameters investigated (distillation curves, density, Reid vapor pressure) showed differentiated behavior, depending on the class of the solvent (oxygenated, light and heavy aliphatic, aromatic) and the quantity added to the gasoline. The azeotropic mixtures formed by ethanol and hydrocarbons showed a strong influence on the behavior of the distillation curves and the location of the point of a sudden change in temperature was shown to be a possible way to detect adulterations and determine the quantity of solvent added to the gasoline.

Keywords: Gasoline adulteration; Azeotropic mixtures; Gasoline properties

1. Introduction

Gasoline is a mixture of several volatile petroleum hydrocarbons with between 4 and 12 carbon atoms, whose distillation range varies between 30 and 225 °C under atmospheric pressure [1]. The hydrocarbons present in gasoline may be classified, basically, into four classes: paraffins (normal and branched), cycloparaffins, olefins and aromatics. The exact composition of gasoline depends on the nature of the petroleum from which it originates (API degree, chemical composition), the process through which the gasoline is obtained (distillation, alkylation, hydrocracking, catalytic cracking, etc.), the end use for which it is produced (automotive competitions, engine performance tests, use in common vehicles), and the legislation in place at the location of production and distribution (contents of benzene, sulfur, lead, etc.).

The refinery gasoline produced in Brazil is called commonly "Type A" gasoline. Brazilian law [2] establishes that the commercial gasoline sold in gas stations in Brazil must be a mixture of anhydrous ethyl alcohol (ethanol) and refinery gasoline. The sale of type A gasoline is not permitted in gas stations in Brazil.

Ethanol is a biofuel with antiknocking properties, which is added to gasoline to replace MTBE (methyl *tert*-butyl ether) and tetra-ethyl lead, and is produced on a large scale in Brazil from the fermentation of sugar cane in several distilleries around the country.

The content of ethanol added to Brazilian gasoline is defined and regulated by the Ministry of Agriculture, Livestock and Provisions of Brazil [2]. This ethanol content can oscillate between 20% and 25%, ±1% by volume, according to the availability of the raw material for its production and oscillations in the price of alcohol and sugar on the internal and external markets. An overview of the Brazilian ethanol program can be found in the literature [3–5].

The practice of commercial gasoline adulteration in Brazil began with the opening of the market to the fuel sector,

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after almost half a century of monopoly of the government, and was worsened by the reduction in hydrous and anhydrous alcohol subsidies and by the liberation of the importation of solvents, making their costs much lower than those of gasoline [6]. Furthermore, the high rate of taxes on gasoline, representing around 50% of its cost [7] to the end consumer, contributes to the high occurrence of this kind of fraud.The properties used for the quality control of commercial automotive gasoline are specified in Brazil by the National Agency for Petroleum, Natural Gas and Biofuels (*Agência Nacional de Petróleo, Gás Natural e Biocombustíveis – ANP*) through the Directive PANP-309 [7] and is showed in Table 1. Of the cases of non-compliance with this directive registered in Brazil in 2007 for commercial gasoline, on average, 44% of cases occurred due to the distillation curve not being within the specifications, 37% for an inappropriate content of ethanol and 10% for an octane number outside the limits [8]. Due to the variability of the chemical composition of gasoline, its quality is always specified through minimum values, maximum values or ranges for the different properties, with the exception of ethanol content which is fixed.

The solvents used in the adulteration of commercial gasoline can be classified as oxygenated, aromatic, and light and heavy aliphatic hydrocarbons, the majority of these compounds being natural constituents of gasoline [1]. The solvents most used in the adulteration of gasoline include diesel oil, kerosene and refined petrochemicals, anhydrous alcohol in excess [9,10], toluene, xylene and hexane [6].

The detection of fraud is difficult when the adulteration involves the addition of a hydrocarbon, or their mixtures, normally present in the composition of gasoline. In these cases, if the quantity of adulterant added is not very high, not even the most sophisticated techniques such as chromatography can be used to qualify the added compound, let alone quantify it, since there is no gasoline considered as the "standard" with which to compare the data. There are cases in which the addition of solvents to commercial gasoline are not detected by the quality tests of PANP-309 [1,9,11], this addition even leading to an improvement in some characteristics, such as octane rating. This occur because the specifications given in PANP-309 were adopted, mostly, based on good engine functioning, not having as a main objective the identification of possible illegal additions of solvents to gasoline. Nevertheless, the action is no less illegal, since it can lead to tax fraud, an increase in environmental pollution, and reduce the performance and durability of the engine [1,9,12], besides leading to unfair competition between gasoline stations.

The addition of solvents changes the original composition of the fuel, affecting its physicochemical properties in different ways [13]. Distillation curves, vapor pressure and octane rating are properties closely related to the fuel composition and the characteristics of its components [11,14,15].

In this study, an investigation was carried out into distillation curves, density and Reid vapor pressure of commercial gasoline after the addition of different solvents in

Table 1
Brazilian specifications for type A gasoline and for commercial gasoline

Parameter	Type A gasoline	Commercial gasoline	Method ABNT[a]	Method ASTM
Color and aspect	Colorless or light yellow and limpid, without impurities.		Visual inspection	Visual inspection
Ethanol (% v/v)	Exempt	25.0 ± 1[b]	NBR 13992	–
Density at 20 °C (kg/m^3)	Not specified	Not specified	NBR 7148 NBR 14065	D 1298, D 4052
Distillation temperatures (°C)			NBR 9619	D 86
10% Evaporated (max.)	65.0	65.0		
50% Evaporated (max.)	120.0	80.0		
90% Evaporated	155.0–190.0	145.0–190.0		
Final boiling point (max.)	220.0	220.0		
Residue (% v/v, max.)	2.0	2.0		
Motor octane number (MON) (min.)	82.0[c]	82.0	MB 457	D 2700
Antiknock index (RON+MON)/2 (min.)	87.0[c]	87.0	MB 457	D 2699, D 2700
Vapor Pressure at 37.8 °C (kPa)	45.0–62.0	69.0 max.	NBR 4149 NBR 14156	D 4953, D 5190, D 5191, D 5482
Gum (mg/100 ml, max.)	5	5	NBR 14525	D 381
Induction period at 100 °C (min.)	360[c]	360	NBR 14478	D 525
Cupper corrosion at 50 °C, 3 h (max.)	1	1	NBR 14359	D 130
Sulfur (% w/w, max.)	0.12	0.10	NBR 6563 NBR 14533	D 1266, D 2622, D 3120, D 4294, D 5453
Benzene (% v/v, max.)	1.2	1.0	–	D 3606, D 5443, D 6277
Lead (g/L, max.)	0.005	0.005	–	D 3237
Hydrocarbons (% v/v)			MB 424	D 1319
Aromatics (max.)	57	45		
Olefins (max.)	38	30		

[a] ABNT: Technical Regulations of the Brazilian Association.
[b] Value currently used in Brazil, Ref. [2].
[c] Value obtained for a mixture (v/v) of 75% refinery gasoline and 25% ethanol.

various proportions, using the limits given in the ANP specification (PANP-309/2001) for the quality of gasoline as a reference. Even without the exact composition of the adulterant and the gasoline it is shown that each solvent has an effect on the physicochemical properties of gasoline of specific magnitude and behavior, depending on the concentration and chemical nature of the solvent (oxygenated, aliphatic, aromatic). It is here proposed to verify which of the properties studied are the most affected by the addition of solvents along with the capacity of detection of these adulterations through the respective quality limits. Particular emphasis is given to the distillation curves of the gasoline samples, including the study of the azeotropic phenomenon which occurs through the mixture of hydrocarbons with oxygenated compounds.

2. Experimental procedure

2.1. Materials

The solvents used in this study were: anhydrous ethanol, diesel, white spirit and alkylbenzene (AB9), the latter two being commercial solvents. These solvents were selected as representative of the main hydrocarbon classes used in gasoline adulteration, oxygenated, heavy and light aliphatic, and aromatic, respectively.

Diesel is composed of hydrocarbons with between 8 and 28 carbon atoms [16]. It is widely used in Brazil as a fuel in heavy transport vehicles like trucks and buses. The diesel sample used in this study is in accordance with the criteria for quality and atmospheric emissions in use in Brazil (Technical Regulation no. 2/2006) [7]. White spirit is a mixture of aliphatic hydrocarbons with between 8 and 16 carbon atoms of petrochemical origin. Alkylbenzene (AB9) is also a petrochemical solvent and is composed of a mixture of aromatic hydrocarbons, containing mainly trimethylbenzene (around 54% w/w on average) and ethyltoluene (around 30% w/w on average) [17].

The chemical composition of the specified (PANP-309) type A gasoline, used as a reference base, are shown in Table 2. Properties for type A gasoline, ethanol and the other solvents are shown in Table 3. A complete description of the anhydrous ethanol used as a gasoline additive in Brazil can be found in ANP Resolution 36/2005 [7].

The type A gasoline, ethanol and diesel were donated by TEBRAS (PETROBRAS Gasoline Distribution Center for the Central West Region). The white spirit and AB9 were acquired from commercial suppliers. Four kinds of mixtures were prepared, whose compositions are shown in Table 4.

In case 1, each of the mixtures has different quantities of ethanol, it being that in this case, mixture number 5 represents the composition of commercial gasoline currently sold in Brazil, in accordance with the PANP-309 specifications. The study of the adulterations in case 1 is of great importance because: (1) in Brazil, most adulteration cases occur through the excessive addition of ethanol, in quantities higher than 25% v/v [7]; (2) it allows the isolated study of the addition of a substance which does not form part of the original gasoline composition and which has a great influence on the physicochemical properties of gasoline.

In cases 2–4, each of the mixtures prepared contains a final constant content of 25% v/v of ethyl alcohol. These cases simulate situations of gasoline adulteration with three different solvents (white spirit, AB9 or diesel) in which the fraudster is concerned with maintaining the final volume of alcohol in the mixture equal to the value required by Brazilian legislation (PANP-309). That is, it simulates a situa-

Table 2
Chemical composition of type A gasoline

Compounds (% v/v)	Type A gasoline	Method
Anhydrous ethyl alcohol	0.0	ABNT NBR 13992
Benzene	0.89	ASTM D 3606, D 5443, D 6277
Olefins	21.3	ASTM D 1319
Saturates	57.7	ASTM D 1319
Aromatics	19.2	ASTM D 1319
Xylenes	4.6	ASTM D 1319
Toluene	3.0	ASTM D 1319

Table 3
Properties of the gasoline and solvents

Properties	Type A gasoline	Ethanol	White spirit	AB9	Diesel	ASTM method
Chemical formula	C_4 to C_{12} compounds	C_2H_5OH	C_8 to C_{16} aliphatic compounds	C_9H_{12} aromatic compounds	C_8 to C_{28} aliphatic compounds	–
Oxygen contend (% w/w)	0.0	34.8	0.0	0.0	0.0	–
Dist. temp. (°C)	35.8–191.4	78.4[a]	151.7–211.5	155.1–174.4	127.5–381.2	D 86
RON	92.9	132[b]	–	–	–	D 2699
MON	81.0	106[b]	–	–	–	D 2700
RVP at 37.8 °C (kPa)	61.7	14.2	3.9	0.4	3.0[c]	D 5191
Density (g/cm^3, 20 °C)	0.7261	0.7931	0.7910	0.8732	0.8598	D 4052
Cetane number	–	–	–	–	44.4	D 613

[a] Ref. [18].
[b] Typical values, Ref. [19].
[c] Ref. [20].

Table 4
Cases studied and composition of mixtures prepared

Mixture	Case 1		Case 2		Case 3		Case 4	
	Type A gasoline and ethanol		Type A gasoline, 25% v/v of ethanol and white spirit		Type A gasoline, 25% v/v of ethanol and AB9		Type A gasoline, 25% v/v of ethanol and diesel	
	Ethanol (% v/v)	Gas. A (% v/v)	White spirit (% v/v)	Gas. A (% v/v)	AB9 (% v/v)	Gas. A (% v/v)	Diesel (% v/v)	Gas. A (% v/v)
1	5	95	5	70	5	70	1	74
2	10	90	10	65	10	65	2	73
3	15	85	15	60	15	60	3	72
4	20	80	20	55	20	55	4	71
5	25	75	25	50	25	50	5	70
6	30	70	30	45	30	45	6	69
7	35	65	40	35	40	35	7	68
8	40	60	50	25	50	25	8	67
9	45	55	–	–	–	–	9	66
10	50	50	–	–	–	–	10	65
11	60	40	–	–	–	–	15	60
12	70	30	–	–	–	–	20	55
13	80	20	–	–	–	–	25	50
14	90	10	–	–	–	–	30	45
15	–	–	–	–	–	–	40	35
16	–	–	–	–	–	–	50	25

tion in which there has been a correction of the ethanol content in the mixture after the solvent addition. Through the study of the adulterations of cases 2–4 it is possible to evaluate the effect which the increase in the quantity of solvent (white spirit, AB9 or diesel) has on the physicochemical properties without the influence of the variation of ethanol content, since this is constant and equal in all mixtures prepared.

2.2. Methods

The physicochemical tests carried out were: distillation curves, density and Reid Vapor Pressure (RVP). The type A gasoline samples, ethanol and the other solvents were previously cooled to a temperature of 0 °C which was held for a minimum period of 24 h. In a flask, immersed in ice and salt, volumetric fractions of each component were added with the aid of volumetric pipettes, in a volume sufficient for the subsequent carrying out of all the tests. The sample flasks were removed briefly from refrigeration, one at a time, only for the additions, the flask being kept closed between these periods. This procedure aims to minimize losses through evaporation and handling errors associated with making the additions. The vapor pressure measurements were carried out first, in triplicate, using closed flasks immersed in ice and salt. The distillation tests were then carried out, in duplicate, and the density was determined in triplicate. The determination of the distillation curves was carried out with a Walter Herzog automatic atmospheric distiller, according to the norm ASTM D86 [21]. For the vapor pressure test a Grabner Instruments Minivap CCA-VPS was used, which provides the Reid vapor pressure (ASTM D5191 [22]). The density was determined with the aid of an Anton Paar DMA 4500 automatic densitometer (ASTM D4052 [23]). All of the tests were carried out at the Reference Laboratory for Fuel Quality – CEPAT – of the National Agency for Petroleum, Natural Gas and Biofuels (ANP).

3. Results and discussion

For the analysis of the results, it should be considered that: in case 1, the content of 0% solvent corresponds to type A gasoline; in cases 2–4, the content of 0% solvent corresponds to the commercial gasoline (25% v/v of ethanol and 75% v/v of gasoline *A*) normally sold at gas stations in Brazil. In all cases, a content of 100% solvent corresponds to pure solvents.

Fig. 1 shows the distillation curves for the type A gasoline and the pure solvents, these curves are used as a reference in the analysis of the effect of the addition of solvents to gasoline. Ethanol, being a pure compound, has a single distillation point, unlike the other compounds which are mixtures of hydrocarbons. The endpoint of the diesel distillation occurred at 90% of volume vaporized since the FBP is not monitored by ANP and to preserve the equipment, given the high FBP temperature.

Figs. 2–6 show the distillation curves of cases 1–4 in Table 4, respectively, where the temperature in °C, the content of added solvent and the volume percentage distilled are shown. The data of case 1 were separated into Figs. 2 and 3 in order to better view them.

The distillation curves of case 1 are clearly differentiated from the other cases, due to the temperature at which the pure solvent is distilled, as can be seen in Figs 2 and 3. Ethanol additions tend to decrease the distillation temperature of the mixture as a whole, unlike diesel, white spirit and AB9.

The distillation curve of type A gasoline shows a continuous and smooth increase in temperature as the distillation process progresses. With the addition of 5% v/v of ethanol

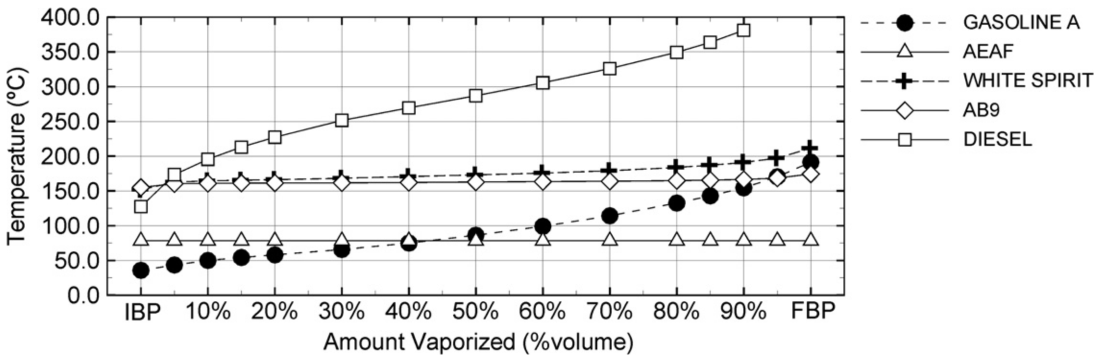

Fig. 1. Distillation curves of solvents compared to type A gasoline.

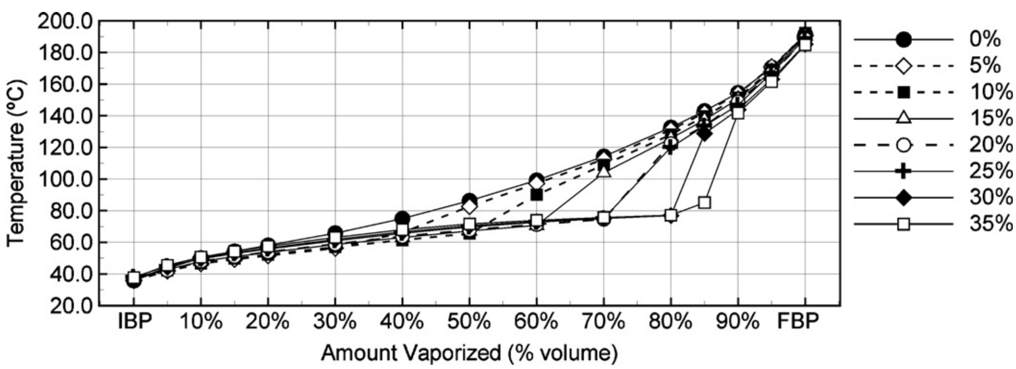

Fig. 2. Distillation curves for type A gasoline–ethanol mixtures from 0% to 35% of ethanol (case 1).

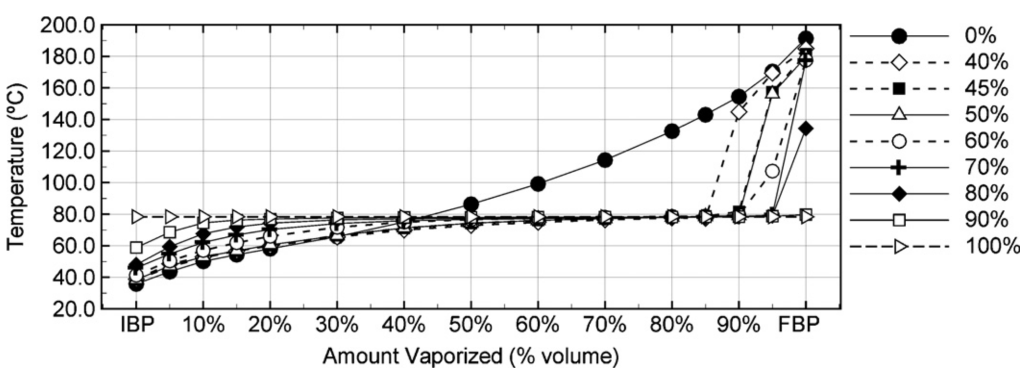

Fig. 3. Distillation curves for type A gasoline–ethanol mixtures from 40% to 100% of ethanol (case 1).

to the gasoline A (case 1) a sudden change in temperature can be observed through a simple visual inspection, which occurred after 40% distillation of the initial volume and from this point onward, 40% vaporized, the distillation curve becomes continuous and smooth again. With the addition of 10% v/v of ethanol to the gasoline *A*, also in Fig. 2, the same sudden change occurred, however, after a distillation of 50% of the initial volume. With the consecutive increase in the proportion of ethanol in the mixture, the location of this sudden increase in temperature tends to occur closer and closer to the final percentage of distillation, as can be seen in Figs. 2 and 3.

The same type of behavior of the distillation curve can also be observed in the mixtures of case 2, with the solvent white spirit, Fig. 4. The mixture with 0% of solvent (25% v/v of ethanol and 75% v/v of type A gasoline) showed a sudden change in temperature after a distillation of 70% of the initial volume. Addition of up to 10% v/v of white spirit did not result in significant changes in the shape of this curve. When the proportion of this solvent reached 15% v/v, the inflexion point dropped to 60% of volume distilled. With an increase in the proportion of white spirit in the mixture, this point shifted to the beginning of the distillation, close to the initial distilled percentages.

The mixtures prepared with white spirit and AB9 (Figs. 4 and 5) distill in a very similar way, despite the former being an aliphatic solvent, and the latter an aromatic solvent. The curves of Fig. 6 (case 4) reveal that the diesel did not affect, in a significant way, the distillation temperature of up to 70% of the vaporized volume for additions of up to 10% of diesel. On the other hand, the FBP was strongly influenced by the addition of diesel, this parameter being the strongest indication of the presence of this substance in gasoline.

In case 4, mixtures with additions of diesel in increments of 1–10% v/v content were prepared since it was noted, during the experiments, that small additions of the compound greatly affect the properties here studied. This affect was noted mainly in the distillation curves, thus verifying that it was not possible for the equipment used to distill mixtures with quantities of diesel above 10%. This occurred due to the very high FBP temperature being very different to that of the gasoline distillation setup.

Mixtures of hydrocarbons with oxygenated compounds, such as ethanol, are far from ideal [11,24,25] and form azeotropic mixtures [26,27]. This phenomenon creates a transition region between the azeotropic mixture and the remaining hydrocarbon mixture, easily identified by a sudden increase in temperature during the distillation, as observed in Figs. 2–6. This phenomenon is well known for bicomponent mixtures of ethanol and hydrocarbons, but remains a challenge when dealing with mixtures containing hundreds of components, as in the case of gasoline. This is of fundamental importance in relation to gasoline adulteration through the excessive addition of ethanol. A better understanding of this phenomenon could aid in the establishment of more appropriate ranges for the quality parameters in order to detect adulterations with greater efficiency.

In Fig. 7 the location of the inflexion points of the distillation curves observed for cases 1–3 are shown. Case 4 is not shown in this figure since the point of sudden temperature change remained at 70% of the vaporized volume for the additions of up to 10% v/v of diesel.

On analyzing of Fig. 7 it is clear that the point on the curve where the sudden increase in temperature occurs depends on the quantity and type of the solvent in the mixture. In case 1, the consecutive additions of ethanol make the inflexion point of the distillation curve tend to be closer and closer to the FBP, in contrast to that which occurs in cases 2 and 3. In case 1 it was not possible to identify the point where the sudden change in temperature occurred for an addition of 90% v/v of ethanol, since the distillation curve became less steep and without inflexion. Also in case 1, for additions of ethanol greater than 25%, the curve formed by the inflexion points assumes values distinct from those of cases 2 and 3, as can be seen in Fig. 7. Such a correlation has never been previously reported in the literature. Thus, the study of azeotropic mixtures formed by ethanol and hydrocarbons allowed the differentiation of the adulterations of cases 2 and 3, from those of case 1,

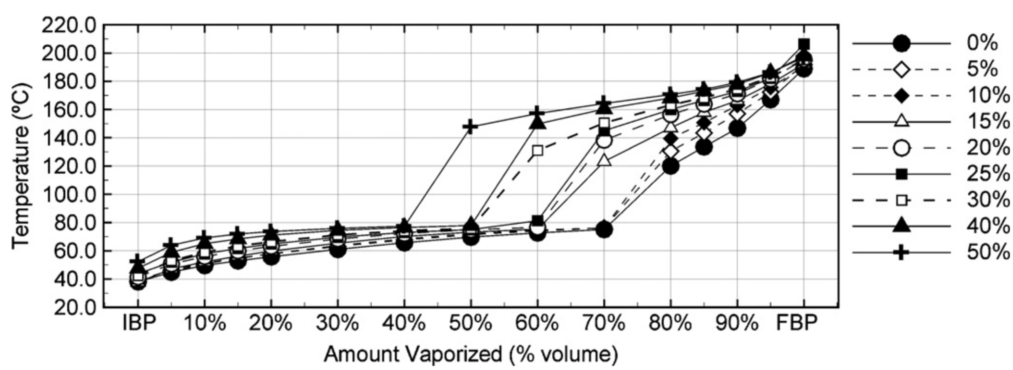

Fig. 4. Distillation curves for gasoline–white spirit mixtures (case 2).

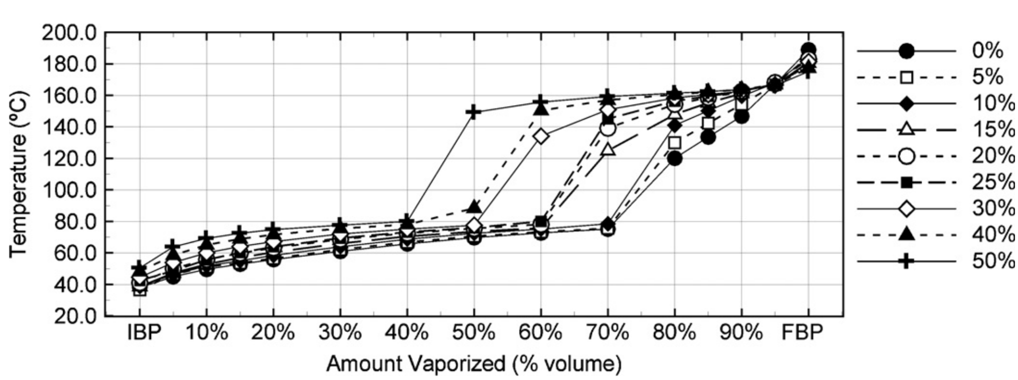

Fig. 5. Distillation curves for gasoline–AB9 mixtures (case 3).

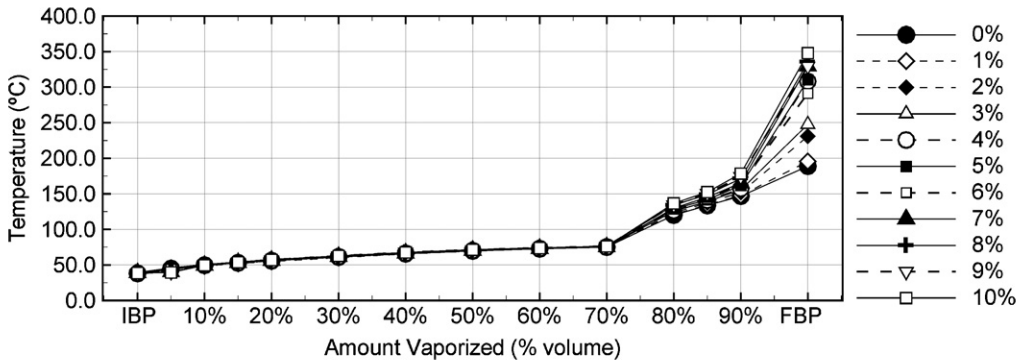

Fig. 6. Distillation curves for gasoline–diesel mixtures (case 4).

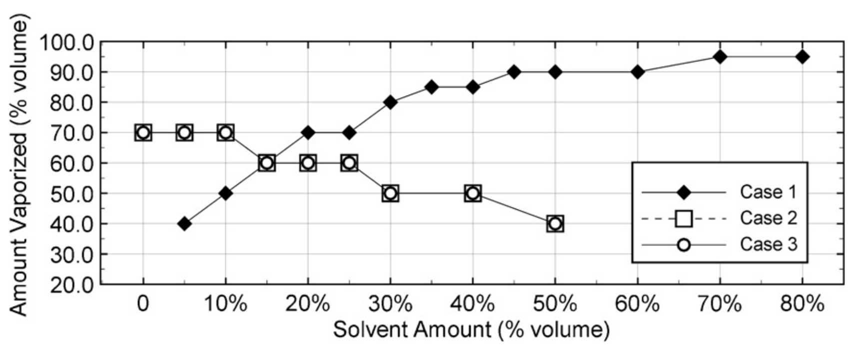

Fig. 7. Inflexion points of distillation curves (cases 1–3).

besides enabling the estimation of the quantity of solvent added to the gasoline mixture. This indicates that the azeotropic phenomenon can be used as a useful tool to aid in the identification and detection of adulterated gasoline, requiring only a qualitative analysis of the distillation curve, which is already collected in the fuel quality control laboratories, thus not incurring any extra costs.

The Brazilian legislation (PANP-309) specifies limit values for the distillation temperatures of the volumetric fractions of 10%, 50%, 90%, and the FBP, as shown in Table 1. The number of points specified is low when compared to a complete distillation curve. This limitation in the number of points may lead to the non-detection of adulterating compounds added to the gasoline [10]. This can also be confirmed in this study. Table 5 provides the values observed for cases 1–4 which surpassed the limits imposed regarding the distillation curve, along with the quantity of solvent present in these mixtures. These data can be observed in the Supplementary Material. In general, it can be noted that these limits only detect adulteration with large quantities of solvent, more than 30% v/v, which is of great concern.

The distillation point that corresponds to 10% of the evaporated volume is related to the engine ignition. The addition of a solvent that increases the volatility of the gasoline can cause the formation of vapor bubbles in the fuel,

Table 5
Solvent content added to gasoline and the corresponding distillation temperature at which the limits established by Brazilian regulations for the quality of commercial gasoline are exceeded

Observed values	10% Evaporated		50% Evaporated		90% Evaporated (temp. max.)		90% Evaporated (temp. min.)		FBP	
	Temp. (°C)	% v/v solvent	Temp. (°C)	% v/v solvent	Temp. (°C)	% v/v solvent	Temp. (°C)	% v/v solvent	Temp. (°C)	% v/v solvent
Case 1	67.6	80.0	82.7	5.0	–	–	143.8	30.0	–	–
Case 2	69.1	50.0	147.7	50.0	–	–	–	–	–	–
Case 3	65.1	40.0	88.4	40.0	–	–	–	–	–	–
Case 4	–	–	–	–	–	–	–	–	231.0	2.0
Limits[a]	65 °C (max.)		80 °C (max.)		190 °C (max.)		145 °C (min.)		220 °C (max.)	

[a] Temperature limits of distillation curves specified by Brazilian legislation (PANP-309).

interrupting the flow and resulting in the engine stalling. A heavier solvent would inhibit the vehicle ignition by decreasing the gasoline volatility. The maximum limit of 65 °C for 10% evaporated was exceeded only for great additions of the tested solvents: 80% of ethanol with 67.6 °C, 50% of white spirit with 69.1 °C and 40% of AB9 with 65.1 °C, as can be seen in Table 5. In case 1, this limit was only exceeded when the properties of the pure ethanol became much more predominant in the mixture. In cases 2 and 3, this limit was exceeded with lower percentages of solvent due to the low volatility of the pure solvents, given by the RVP values for white spirit (3.9 kPa) and AB9 (0.4 kPa). In case 4, the diesel contents tested did not affect the distillation of the mixture for up to 70% of distilled volume (Fig. 6), and therefore the limit of 10% was not disrespected in this case.

The maximum limit of 80 °C for 50% of evaporated volume was exceeded for the mixtures with 50% of white spirit (147.7 °C) and 40% of AB9 (88.4 °C) and for the mixture of type A gasoline with 5% of ethanol (82.7 °C). However, it should be noted that in Brazil the gasoline sold at gas stations must be a mixture of type A gasoline with ethanol between 20% and 25%, ±1% by volume. Only additions of ethanol greater than these values characterize an adulteration with ethanol. The 50% distilled limit is monitored by its relation with the heating and accelerating of the engine and with the fuel economy.

The maximum limits established for the temperature of a 90% evaporated volume and of the FBP aim to control the quantity of high boiling point compounds and to avoid the formation of carbon deposits inside the engine [13]. The maximum limit of 190 °C for the 90% evaporated was not exceeded in any of the four cases. The minimum limit of 145 °C was exceeded for additions greater than 30% of ethanol (distillation at 143.8 °C). The limit imposed on the minimum temperature for this distilled percentage aims to inhibit the addition of solvents to commercial gasoline, which was efficient just for the adulterations of case 1, that is, an excess of ethanol.

The maximum limit of 220 °C for FBP was reached only in case 4 with the addition of 2% of diesel leading to an FBP of 231.0 °C. The other cases remained below the limit. Because diesel is composed of hydrocarbons of between 8 and 28 carbon atoms [16], the detection of adulteration by heavy aliphatic solvents, such as diesel, is easier because the mixture contains compounds which are not present in the original gasoline composition [1].

The definition of a limit for the RVP (69.0 kPa), Table 1, is important for fuel safety and performance, being a measure of the tendency to form explosive vapors. Fig. 8 shows the results of the RVP tests carried out for the cases given in Table 4.

The consecutive additions of ethanol to type A gasoline, case 1, caused an increase in the RVP up to a maximum of 68.9 kPa for 10% v/v ethanol, and from then on the pressure decreased and did not exceed the maximum limit of 69.0 kPa given in PANP-309. This figure also shows the RVP data for ethanol, at 37.8 °C, taken from the literature [24], where the similarity of the literature data with those of this study can be observed. In [24] the authors obtained a higher vapor pressure value (71.6 kPa) for an addition of 10% of ethanol, with a decrease in the RVP from this point onward. It is important to note that, although the type A gasoline mixtures used in this study and in the research by Pumphrey et al. are not the same, there were great qualitative and quantitative similarities regarding the RVP in both cases, which shows that it is possible to estimate the behavior of the adulterations, even when the exact composition of the adulterated gasoline is not known. In other studies, the RVP peak was obtained for lower percentages of ethanol addition to gasoline (3–5% vol.) [26,27]. However, in these studies the RVP reference base for the gasoline used was around 80.0 kPa, a much more volatile gasoline than that used in this study, and even so, the qualitative behavior is similar.

The vapor pressure, being an indicator of fuel volatility, is closely related to the molecular interactions of the mixture components [16]. As seen in Fig. 8, the initial increase and subsequent decrease in RVP occurs only in case 1, where the ethanol content is not constant, the point of greatest vapor pressure corresponding to a mixture with 10% ethanol and 90% gasoline. In cases 2–4, the ethanol content is fixed at 25% v/v. This analysis indicates that ethanol is responsible for the increase in vapor pressure as a result of the decrease in molecular interaction between this compound and the other hydrocarbons. Subsequent

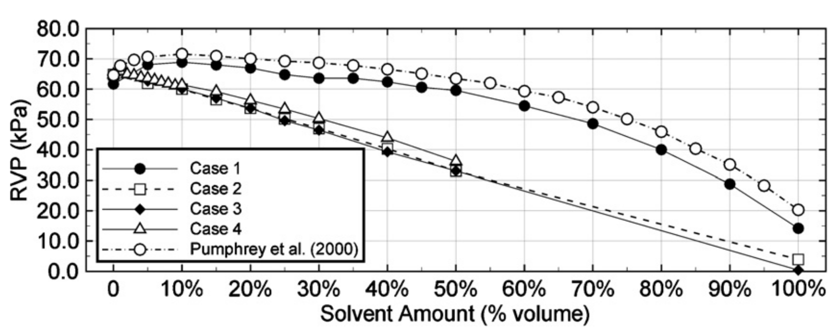

Fig. 8. Reid vapor pressure of gasoline for cases 1–4.

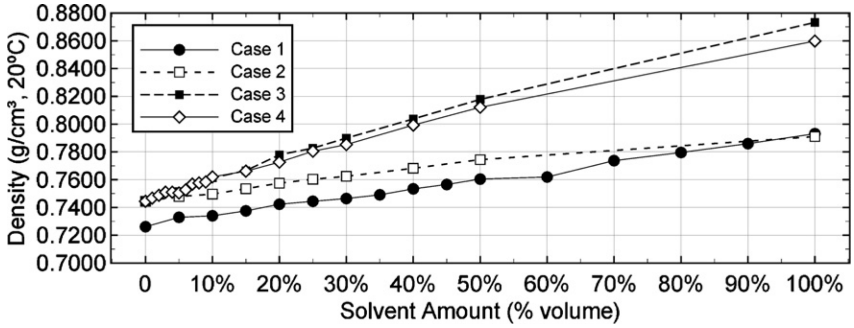

Fig. 9. Gasoline density for cases 1–4.

additions of ethanol cause the RVP to decrease due to the predominance of stronger molecular interactions making the mixture less volatile. Since in cases 2–4 the ethanol content is fixed, there is a linear reduction in the vapor pressure only as a consequence of the addition of less volatile compounds to the gasoline.

Fig. 9 gives the density of the gasoline at 20 °C for all four cases. As can be seen in this figure, the increase in the solvent concentration in the gasoline increases the density of the mixtures, which may be attributed to the higher density of the pure compounds in comparison to the type A gasoline, as shown in Table 3. Also, two distinct behaviors can be observed; for cases 1 and 2 and cases 3 and 4 there is an increase in density with different intensities, which is to be expected, since the density of the solvents ethanol and white spirit are very similar, as are those of AB9 and diesel.

The Brazilian legislation does not specify a limit for the density, however, a high value may indicate the addition of a dense compound such as AB9. Côcco et al. [28] provide a collection of density data for Brazilian commercial gasoline samples, most samples having values between 0.74 and 0.78 g/cm^3. Taking these values as a limit, gasoline samples with AB9 or diesel contents higher than 20% (cases 3 and 4), or ethanol contents above 80% v/v (case 1), would be classified as not conforming to the density limit.

Changes in the density affect the quantity of fuel that is injected into the combustion chamber during each cycle, altering the ideal air-fuel burning ratio. In the case of a gasoline adulterated with a more dense solvent, the burning will be incomplete, resulting in the emission of pollutants to the atmosphere.

4. Conclusions

A study was carried out on the influence of the addition of the solvents anhydrous ethyl alcohol (ethanol), white spirit, alkylbenzene AB9 and diesel on the parameters Reid vapor pressure, density and distillation curves. These solvents represent the four classes of hydrocarbons used in the adulteration of gasoline: oxygenated, light aliphatics, aromatics and heavy aliphatics, respectively. The mixtures with ethanol and type A gasoline give distillation curves that are very different to the other cases. In general, the additions of AB9, diesel and white spirit to gasoline increase the distillation temperature of the mixture, while the ethanol tends to reduce it. Diesel was the compound which was most easily detected, since it results in an increase in the FBP above the limit with the addition of only 2% by volume. On the distillation curves of the four cases studied it was possible to identify the point of a sudden change in temperature, characteristic of azeotropic mixtures formed by ethanol and hydrocarbons, through a simple visual inspection. It was found that the location of these points on the curve depends on the quantity and type of solvent added to the gasoline, which makes possible a distinction between the adulterations of case 1 and cases 2 and 3, besides allowing the estimation of the quantity of solvent added to the mixture. The location of the inflexion point has never been used before to study gasoline adulterations and it was shown to be a tool of great potential in the detection and identification of adulterations through the analysis of the distillation curve. Since it is currently collected in the fuel quality control laboratories, analysis of the distillation curve does not incur extra costs. Further, more in-depth, studies are needed in order to elucidate the formation of these azeotropes. Regarding the density results, the increase in the solvent concentration in the gasoline increases the density of the mixtures. The Brazilian legislation does not specify a limit for the density, however, a high value may indicate the addition of a dense compound such as AB9. Values between 0.74 and 0.78 g/cm^3 are suggested as limits for this parameter. The similarity of the data of this study with those given in the literature for RVP shows that it is possible to estimate the behavior of the adulterations, even when the exact composition of the adulterated gasoline is not known. Although the study deals specifically with the Brazilian problem, this being the greatest consumer market of ethanol for automotive purposes, its reach is much broader and of interest to the international scientific community, since, given the possibility for the use of ethanol in other countries in their automotive fleets, the practice of adulteration may affect new markets which are adopting this technology. Brazil has had the know-how for the use of ethanol since the installation of the PROALCOOL program, and the analysis and improvement of control techniques and practices in the Brazilian market may contribute to the development of energy policies and quality control of fuels in these new markets.

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Appendix A. Supplementary data

Supplementary data associated with this article can be found, in the online version, at doi:10.1016/j.fuel.2007.11.003.

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